

Incompressible fluids II: Stokes

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Fluid mechanics: governing principles

We denote by

ρ = density,

u = velocity,

σ = Cauchy stress tensor,

f = body force.

The **incompressible Navier–Stokes** and **Stokes** systems of equations are derived from the physical conservation law

- ▶ conservation of mass,
- ▶ conservation of momentum.

We have seen that, in particular if $\rho \equiv \text{const}$, conservation of mass yields the **incompressibility constraint**

$$\nabla \cdot u = 0.$$

Conservation of momentum

Momentum density is ρu . Conservation of momentum over ω reads

$$\frac{d}{dt} \int_{\omega} \rho u \, dx + \int_{\partial\omega} \rho u (u \cdot n) \, ds = \int_{\partial\omega} \sigma n \, ds + \int_{\omega} f \, dx.$$

Applying the divergence theorem gives¹

$$\partial_t(\rho u) + \nabla \cdot (\rho u \otimes u) = \nabla \cdot \sigma + f.$$

For an incompressible fluid with constant density, using the identity

$$\nabla \cdot (u \otimes u) = (u \cdot \nabla)u + (\nabla \cdot u)u = (u \cdot \nabla)u,$$

we obtain

$$\rho \partial_t u + \rho (u \cdot \nabla)u = \nabla \cdot \sigma + f.$$

¹The tensor product $u \otimes u$ is the matrix with entries $(u \otimes u)_{ij} := u_i u_j$.

Constitutive equation: Newtonian fluids

As in elasticity, **the system is closed by a constitutive law for the stress tensor.**

For a **Newtonian fluid**,

$$\sigma = -pI + \mu(\nabla u + \nabla u^T) = -pI + 2\mu\varepsilon(u), \quad \varepsilon(u) = \frac{1}{2}(\nabla u + \nabla u^T).$$

Here

- ▶ p is the pressure. $-pI$ represents isotropic stress
- ▶ μ is the dynamic viscosity. The viscous shear stress is proportional to the rate of deformation tensor $\varepsilon(u)$

Substituting into the momentum equation gives

$$\rho \partial_t u + \rho(u \cdot \nabla)u = -\nabla p + 2\nabla \cdot (\mu \varepsilon(u)) + f.$$

From constitutive law to Navier–Stokes

If μ is constant, then

$$2\nabla \cdot (\mu \varepsilon(u)) = \mu \nabla \cdot (\nabla u + \nabla u^T).$$

Using

$$\nabla \cdot (\nabla u + \nabla u^T) = \Delta u + \nabla(\nabla \cdot u),$$

and incompressibility, one gets

$$2\nabla \cdot (\mu \varepsilon(u)) = \mu \Delta u.$$

Therefore

$$\rho \partial_t u + \rho(u \cdot \nabla)u - \mu \Delta u + \nabla p = f.$$

Dividing by ρ and writing $\nu = \mu/\rho$,

$$\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f.$$

Boundary conditions and pressure normalization

We thus obtain the [Navier-Stokes system](#):

$$\begin{cases} \partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f, \\ \nabla \cdot u = 0. \end{cases}$$

which must be complemented with suitable boundary conditions, eg.:

- ▶ slip wall: $u \cdot n = 0$,
- ▶ no-slip wall: $u = g_D$,
- ▶ stress-free outflow: $\sigma n = 0$,
- ▶ do-nothing outflow: $n \cdot \nabla u - pn = 0$.

Since the pressure is defined only up to an additive constant, one often imposes

$$\int_{\Omega} p \, dx = 0, \quad \text{i.e. } p \in L_0^2(\Omega) := \left\{ q \in L^2(\Omega) : \int_{\Omega} q = 0 \right\}.$$

The Stokes system

For systems that are easier to study, and relevant in laminar-flow regimes, one simplifies Navier–Stokes by omitting convection. If, moreover, one assumes steady state, we obtain the **Stokes system**

$$\begin{cases} -2\mu \nabla \cdot \varepsilon(u) + \nabla p = f & \text{in } \Omega, \\ \nabla \cdot u = 0 & \text{in } \Omega, \\ u = g_D & \text{on } \partial\Omega. \end{cases}$$

As we assume μ is constant and $\nabla \cdot u = 0$, this is equivalent to

$$\begin{cases} -\mu \Delta u + \nabla p = f & \text{in } \Omega, \\ \nabla \cdot u = 0 & \text{in } \Omega, \\ u = g_D & \text{on } \partial\Omega. \end{cases}$$

Why Stokes is the relevant model here

The Stokes system is the prototypical incompressible saddle-point problem. It already contains the key analytical and numerical difficulty:

- ▶ the velocity is constrained by incompressibility,
- ▶ the pressure acts as a Lagrange multiplier.

So, even before treating the full Navier–Stokes system, one must understand:

- ▶ the divergence constraint,
- ▶ the zero-mean pressure condition,
- ▶ inf–sup stable discretizations.

Stokes system: weak form

From incompressible Stokes with homogeneous Dirichlet b.c.

$$\begin{cases} -2\mu \nabla \cdot \varepsilon(u) + \nabla p = f & \text{in } \Omega, \\ \nabla \cdot u = 0 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

In mixed form: find

$$u \in V := [H_0^1(\Omega)]^d, \quad p \in Q := L_0^2(\Omega),$$

such that

$$\begin{cases} a(u, v) + b(v, p) = (f, v) & \forall v \in V, \\ b(u, q) = 0 & \forall q \in Q, \end{cases}$$

where

$$a(u, v) = 2\mu \int_{\Omega} \varepsilon(u) : \varepsilon(v), \quad b(v, q) = - \int_{\Omega} q \nabla \cdot v.$$

Well-posedness of continuous Stokes

Korn's inequality gives

$$\|\nabla v\|_{L^2(\Omega)} \leq C \|\varepsilon(v)\|_{L^2(\Omega)} \quad \forall v \in [H_0^1(\Omega)]^d.$$

Hence

$$a(v, v) = 2\mu \|\varepsilon(v)\|_{L^2(\Omega)}^2 \gtrsim \|v\|_{H^1(\Omega)}^2,$$

so a is coercive on V and in particular on the divergence-free subspace Z .
The divergence operator is surjective

$$\nabla \cdot : [H_0^1(\Omega)]^d \rightarrow L_0^2(\Omega),$$

so the inf-sup condition holds. Therefore the continuous Stokes problem is well posed, with the solution satisfying

$$\|u\|_V + \|p\|_Q \leq C \|f\|_{H^{-1}(\Omega)}.$$

Discrete Stokes and divergence operator

For **Stokes**, we require discretisations **conforming in H^1** , and not just in $H(\text{div})$ as in the Darcy case. **For the natural spaces pairs, the divergence is not surjective, hence inf-sup is non-trivial.**

Let $V_h \subset V$ and $Q_h \subset Q$ be conforming discrete spaces.

Define the discrete divergence operator by

$$B_h = -\nabla_h : V_h \rightarrow Q_h, \quad (B_h u_h, q_h)_Q = b(u_h, q_h) = - \int_{\Omega} q \nabla \cdot v.$$

Hence,

$$\nabla_h u_h = \Pi_h(\nabla \cdot u_h),$$

where Π_h denotes the L^2 projection onto Q_h .

If $\nabla \cdot u_h \in Q_h$, then $\nabla_h \cdot V_h = \nabla \cdot V_h$, but this is not true in general. In this case, the discrete solution need not be exactly divergence free.

Why bad pairs fail: locking vs spurious pressures

The incompressibility constraint is imposed through the discrete divergence operator

$$B_h : V_h \rightarrow Q_h, \quad (B_h v_h, q_h) = b(v_h, q_h).$$

The discrete velocity must satisfy $B_h u_h = 0$, so the relevant space is

$$Z_h := \ker B_h = \{v_h \in V_h : b(v_h, q_h) = 0 \ \forall q_h \in Q_h\}.$$

Two pathologies may occur if the pair (V_h, Q_h) is not compatible:

- ▶ **Locking:** Z_h is too small, so the discrete velocity space cannot represent incompressible flows.
- ▶ **Spurious pressure modes:** there exists $q_h \neq 0$ such that

$$b(v_h, q_h) = 0 \quad \forall v_h \in V_h,$$

so some pressures are invisible to the scheme.

The discrete inf-sup condition prevents both phenomena (pressures cannot be invisible, the incompressibility constraint is not over-restrictive)...

Discrete inf-sup

Given that a is coercive in V , the only issue is the satisfaction of the inf-sup condition:

Theorem (well-posedness of discrete formulation)

If $V_h \times Q_h$ satisfies the discrete inf-sup condition

$$\inf_{q_h \in Q_h \setminus \{0\}} \sup_{v_h \in V_h \setminus \{0\}} \frac{\int_{\Omega} q_h \nabla \cdot v_h \, dx}{\|q_h\|_{L^2(\Omega)} \|v_h\|_{H^1(\Omega)}} \geq \beta_h \geq \beta > 0,$$

then the discrete Stokes problem: find $(u_h, p_h) \in V_h \times Q_h$ s.t.

$$\begin{cases} a(u_h, v) + b(v, p_h) &= (f, v) & \forall v \in V_h, \\ b(u_h, q) &= 0 & \forall q \in Q_h, \end{cases}$$

has a unique solution and²

$$\|u - u_h\|_V + \|p - p_h\|_Q \leq C \left[\inf_{v_h \in V_h} \|u - v_h\|_V + \inf_{q_h \in Q_h} \|p - q_h\|_Q \right].$$

²This is the mixed analogue of Céa's lemma. Note that velocity error depends on pressure error.

Example: P_1 - P_0 and locking

Take

$$V_h = \{v_h \in [H_0^1(\Omega)]^d : v_h|_K \in [\mathbb{P}_1(K)]^d\}, \quad Q_h = \{q_h \in L_0^2(\Omega) : q_h|_K \in \mathbb{P}_0(K)\}.$$

Define the discrete divergence-free space

$$Z_h := \{v_h \in V_h : b(v_h, q_h) = 0 \ \forall q_h \in Q_h\}.$$

Then the first equation is equivalent to finding $u_h \in Z_h$ such that

$$a(u_h, v_h) = (f, v_h) \quad \forall v_h \in Z_h.$$

However, it may even happen that

$$Z_h = \{0\},$$

for dimensional reasons on certain meshes (let alone satisfying the surjectivity condition). This is an extreme case of **locking phenomenon**: **the velocity space is not rich enough to satisfy the discrete divergence constraint.**

Example: equal-order pairs and spurious pressures

Consider the continuous equal-order pair

$$V_h = \{v_h \in [H_0^1(\Omega)]^d : v_h|_K \in [\mathbb{P}_1(K)]^d\}, \quad Q_h = \{q_h \in H_0^1(\Omega) : q_h|_K \in \mathbb{P}_0(K)\}.$$

Also this pair does not satisfy inf-sup in general.

For instance, P_1 - P_1 may fail because the kernel of the discrete gradient is nontrivial, leading to

spurious pressure modes

rather than locking.

This shows that the issue is not merely approximation order: it is the compatibility of the two spaces.

Taylor–Hood elements

A standard inf–sup stable conforming pair is the Taylor–Hood element:

$$V_h = \{v_h \in [H_0^1(\Omega)]^d : v_h|_K \in [\mathbb{P}_2(K)]^d\}, \quad Q_h = \{q_h \in H_0^1(\Omega) : q_h|_K \in \mathbb{P}_1(K)\}.$$

More generally, the pair

$$(\mathbb{P}_k, \mathbb{P}_{k-1}), \quad k \geq 2,$$

is inf–sup stable on suitable meshes: there exists $\beta > 0$ independent of h such that

$$\sup_{v_h \in V_h \setminus \{0\}} \frac{\int_{\Omega} q_h \nabla \cdot v_h \, dx}{\|v_h\|_{H^1(\Omega)}} \geq \beta \|q_h\|_{L^2(\Omega)} \quad \forall q_h \in Q_h.$$

Stokes error estimate for Taylor–Hood

For the Taylor–Hood pair $(\mathbb{P}_k, \mathbb{P}_{k-1})$, $k \geq 2$, assuming sufficient regularity of the exact solution,

$$\|u - u_h\|_{H^1(\Omega)} + \|p - p_h\|_{L^2(\Omega)} \leq Ch^k \left(|u|_{H^{k+1}(\Omega)} + |p|_{H^k(\Omega)} \right).$$

In the lowest-order Taylor–Hood case (P_2, P_1) , this reads

$$\|u - u_h\|_{H^1(\Omega)} + \|p - p_h\|_{L^2(\Omega)} \leq Ch^2 \left(|u|_{H^3(\Omega)} + |p|_{H^2(\Omega)} \right).$$